

LAB1 : Centralized Audit Trail Compliance Engine

Introduction

The **Centralized Audit Trail Compliance Engine** is a compliance logging platform designed to securely manage enterprise application audit logs. The system ingests structured audit logs, identifies and masks sensitive Personally Identifiable Information (PII) such as Social Security Numbers and credit card numbers, and stores the protected records with integrity verification. It also allows authorized users to search and filter historical audit records and generate cryptographically verifiable, immutable audit exports for compliance and security auditing.

Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall parse JSON logs and mask PII before storage.	High	Pass: PII is masked before storage. Fail: PII is stored in cleartext.	Protects sensitive data.	Given requirement
FR-002	Functional	The system shall validate incoming JSON audit logs.	High	Pass: Invalid logs are rejected. Fail: Invalid logs are stored.	Ensures valid audit data.	-
FR-003	Functional	The system shall generate a SHA-256 hash for each audit record.	High	Pass: Every record has a SHA-256 hash. Fail: A record has no hash.	Ensures data integrity.	-
FR-004	Functional	The system shall allow authorized users to search and filter audit logs.	High	Pass: Matching records are displayed. Fail: Unauthorized records are displayed.	Supports auditing and investigation.	-
FR-005	Functional	The system shall generate audit exports with integrity information.	High	Pass: Export contains records and hashes. Fail: Records or hashes are missing.	Provides reliable audit evidence.	-
NFR-001	Nonfunctional	The system shall filter 10 million records in under 1 second.	High	Pass: Response time is under 1 second. Fail: Response time exceeds 1 second.	Ensures fast queries.	Given requirement
NFR-002	Nonfunctional	The system shall ensure audit exports are immutable and verifiable.	High	Pass: Modified exports fail verification. Fail: Modified exports pass verification.	Protects audit integrity.	-

Actors

1. Compliance Officer
2. Security Auditor
3. Enterprise Application / Log Source

Use Cases

1. Ingest Audit Logs
2. Validate Audit Logs
3. Mask PII
4. Search & Filter Audit Logs
5. Generate Audit Export
6. Verify Audit Integrity

UML User Case Diagram

